



विशाल जाधव सरांचे
VJTech Academy
Inspiring Your Success...

“UNIT-I Java Language Programs”

VJTech Academy
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Live Class Lectures

Program: WAP to display "Hello World" message on output

```
class VJTech
{
    public static void main(String args[])
    {
        System.out.println("Hello World");
    }
}
```

```
/*
set path="C:\Program Files\Java\jdk-1.8\bin";
1) Java Code Compilation:
=> javac filename.java
2) Java Code Execution:
=> java filename
*/
```

Program: WAP to calculate total and average marks

```
import java.lang.*;
class Result
{
    public static void main(String args[])
    {
        int s1,s2,s3,s4,s5,total;
        float avg;
        s1=88;
        s2=35;
        s3=67;
        s4=89;
        s5=94;
        total=(s1+s2+s3+s4+s5);
        avg=(float)total/5;
        System.out.println("Total Marks:"+total);
        System.out.println("Average Marks:"+avg);
    }
}
```

Program: WAP to demonstrate command line argument

```
class CommandLineArgsDemo
{
    public static void main(String args[])
    {
        int a,b,c;
        a=Integer.parseInt(args[0]);
        b=Integer.parseInt(args[1]);
        c=a+b;
        System.out.println("Addition="+c);
    }
}
```

/*

1) Compilation:

javac filename.java

2) Execution:

java filename 100 200

*/

Program: WAP to demonstrate command line argument

```
class CommandLineArgsDemo1
{
    public static void main(String args[])
    {
        float radius,area;
        radius=Float.parseFloat(args[0]);
        area=(3.14f*radius*radius);
        System.out.println("Area of Circle="+area);
    }
}
```

/*

1) Compilation:

javac filename.java

2) Execution:

java filename 2.2

*/

Program: WAP to demonstrate command line argument

```
class CommandLineArgsDemo2
{
    public static void main(String args[])
    {
        int s1,s2,s3,s4,s5,total;
        float avg;
        s1=Integer.parseInt(args[0]);
        s2=Integer.parseInt(args[1]);
        s3=Integer.parseInt(args[2]);
        s4=Integer.parseInt(args[3]);
        s5=Integer.parseInt(args[4]);
        total=(s1+s2+s3+s4+s5);
        avg=(float)total/5;
        System.out.println("*****");
        System.out.println("***VJTECH ENGG COLLEGE***");
        System.out.println("*****");
        System.out.println("Roll NO: 101010");
        System.out.println("Name: James Gosling");
        System.out.println("Physics Marks:"+s1);
        System.out.println("Chemistry Marks:"+s2);
        System.out.println("English Marks:"+s3);
        System.out.println("Math Marks:"+s4);
        System.out.println("Java Marks:"+s5);
        System.out.println("=====");
        System.out.println("Total Marks:"+total);
        System.out.println("Average Marks:"+avg);
    }
}
```

Program: Scanner class

```
import java.util.*;
class ScannerDemo
{
    public static void main(String args[])
    {
        int a,b,c;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter first number:");
        a=sc.nextInt();
        System.out.println("Enter second number:");
        b=sc.nextInt();
        c=a+b;
        System.out.println("Addition="+c);
    }
}
```

Program: Math class Program

```
class MathMethods
{
    public static void main(String args[])
    {
        int m=10,n=20;
        System.out.println("The minimum Value = "+Math.min(m,n));
        System.out.println("The maximum Value = "+Math.max(m,n));
        System.out.println("Square root of 9 = "+Math.sqrt(9));
        System.out.println("Pow(2,4) = "+Math.pow(2,4));
        System.out.println("exponential of 709.78222656 is "+Math.exp(709.78222656));
        System.out.println("round(200.675) = "+Math.round(200.675));
        System.out.println("round(200.50) = "+Math.round(200.50));
        System.out.println("round(200.20) = "+Math.round(200.20));
        System.out.println("Absolute Value = "+Math.abs(-5944));
    }
}
```

/*

OUTPUT:

The minimum Value = 10

The maximum Value = 20

Square root of 9 = 3.0

Pow(2,4) = 16.0

exponential of 709.78222656 is 1.7968190692375724E308

round(200.675) = 201

round(200.50) = 201

round(200.20) = 200

Absolute Value = 5944

*/

Program: Implicit Type casting

```
class ImplicitTypeCastingDemo
{
    public static void main(String args[])
    {
        int a=70;
        float b;
        b=a;
        System.out.println("Value of int variable a="+a);
        System.out.println("Value of float variable b="+b);
    }
}
```

Program: Explicit Type casting

```
class ExplicitTypeCastingDemo
{
    public static void main(String args[])
    {
        int a=70;
        float b;
        b=(float)a;
        System.out.println("Value of int variable a="+a);
        System.out.println("Value of float variable b="+b);
    }
}
```

Program: instanceof operator

```
import java.util.*;
class InstanceOfOpDemo
{
    public static void main(String args[])
    {
        Scanner sc=new Scanner(System.in);
        if(sc instanceof Scanner)
        {
            System.out.println("sc is an object of Scanner class");
        }
    }
}
```

Write a Java program to check whether two numbers are same or not.

```
import java.util.*;
class IfStatement
{
    public static void main(String args[])
    {
        int a,b;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter Two Integer numbers:");
        a=sc.nextInt();
        b=sc.nextInt();
        if(a==b)
        {
            System.out.println("Both numbers are equal!!!");
        }
    }
}
```


Write a Java program to check whether two numbers are same or not.

```
import java.util.*;
class IfElseStatement
{
    public static void main(String args[])
    {
        int a,b;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter Two Integer numbers:");
        a=sc.nextInt();
        b=sc.nextInt();
        if(a==b)
        {
            System.out.println("Both numbers are equal!!!");
        }
        else
        {
            System.out.println("Both numbers are not equal");
        }
    }
}
```

Write a Java program to check whether number is positive or Negative.

```
import java.util.*;
class PositiveNegativeNo
{
    public static void main(String args[])
    {
        int no;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter Number:");
        no=sc.nextInt();
        if(no!=0)
        {
            if(no>0)
            {
                System.out.println("Positive Number");
            }
            else
            {
                System.out.println("Negative Number");
            }
        }
        else
        {
            System.out.println("Zero is neither positive nor negative");
        }
    }
}
```

/*Write a Java program to generate student marks grade on the basis of following conditions.

marks>=75 - Distinction

marks>=60 - First Class

marks>=40 - Pass

marks<40 - Fail

***/**

```
import java.util.*;
class ElselfLadderDemo
{
    public static void main(String args[])
    {
        int marks;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter your marks:");
        marks=sc.nextInt();
        if(marks>=75)
        {
            System.out.println("You got Distinction!!!");
        }
        else if(marks>=60)
        {
            System.out.println("You got First Class!!!");
        }
        else if(marks>=40)
        {
            System.out.println("You are pass only!!!");
        }
        else
        {
            System.out.println("You are Fail!!!");
        }
    }
}
```

Write a Java program to demonstrate switch case statement

```
import java.util.*;
class SwitchCaseDemo
{
    public static void main(String args[])
    {
        int no;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter number between 1 to 3:");
        no=sc.nextInt();
        switch(no)
        {
            case 1: System.out.println("You have entered ONE");
                    break;
            case 2: System.out.println("You have entered TWO");
                    break;
            case 3: System.out.println("You have entered THREE");
                    break;
            default: System.out.println("Invalid Number");
        }
    }
}
```

Write a Java program to check whether number is EVEN or ODD.

```
import java.util.*;
class EvenOddNProgram
{
    public static void main(String args[])
    {
        int no;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter Number:");
        no=sc.nextInt();
        if(no%2==0)
        {
            System.out.println("Even Number");
        }
        else
        {
            System.out.println("Odd Number");
        }
    }
}
```

For loop.

```
class forloopDemo
{
    public static void main(String args[])
    {
        int i;
        for(i=1;i<=5;i++)
        {
            System.out.println("VJTech Academy");
        }
    }
}
```

//While Loop

```
class whileloopDemo
{
    public static void main(String args[])
    {
        int i=1;
        while(i<=5)
        {
            System.out.println("VJTech Academy");
            i++;
        }
    }
}
```

//Do-while loop

```
class DowhileloopDemo
{
    public static void main(String args[])
    {
        int i=11;
        do
        {
            System.out.println("VJTech Academy");
            i++;
        }while(i<=5);
    }
}
```

//For Each Loop

```
class ForEachLoopDemo
{
    public static void main(String args[])
    {
        int num[]={10,20,30,40,50};
        System.out.println("Your Array Elements are:");
        for(char x:num)
        {
            System.out.println("Value of x :"+x);
        }
    }
}
```

//Constant Variable using final keyword

```
class ConstantVariableDemo
{
    public static void main(String args[])
    {
        final int a=100;
        System.out.println("Value of a="+a);
    }
}
```

Write a Java program to find out the even numbers from 1 to 100 using for loop.

```
class Print1to100EvenNo
{
    public static void main(String args[])
    {
        int i;
        System.out.println("Even Number between 1 to 100");
        for(i=1;i<=100;i++)
        {
            if(i%2==0)
            {
                System.out.println(i);
            }
        }
    }
}
```

Write a Java program to find out the odd numbers from 1 to 100 using for loop.

```
class Print1to100OddNo
{
    public static void main(String args[])
    {
        int i;
        System.out.println("ODD Number between 1 to 100");
        for(i=1;i<=100;i++)
        {
            if(i%2!=0)
            {
                System.out.println(i);
            }
        }
    }
}
```